EEG Stress Analyser

Instruction Manual - band-power stress comparison, live and offline

1. What this program measures

This tool compares EEG band power across up to three recordings of one participant: a resting **Baseline**, a recording taken **Before** a stressor task, and one taken **After** it. It answers three questions: how do all frequency bands differ between any two of the three recordings; how did the alpha/beta ratio change from Before to After; and, treating Baseline as this person's normal level, how much did the stressor move each measure beyond wherever they already sat.

Six bands are reported: delta, theta, alpha, low beta, high beta and gamma, plus the alpha/beta and theta/beta ratios. Two headline numbers appear throughout: **signal-to-noise** style quality checks are not part of this tool - see the EEG Optimizer for that - this tool's headline is the **alpha/beta ratio**, a standard index where a lower value indicates more beta relative to alpha, generally read as more alert or more stressed.

> Runs entirely on the CPU, no graphics card needed. Files never leave the computer. The live page talks only to the EMOTIV Cortex service on this same machine (wss://localhost:6868).

2. Installing

You need **Python 3.10 or newer**, with *Add python.exe to PATH* ticked during installation.

1. Double-click **install.bat** in this folder.
2. Wait for it to finish, then close the window.

```
pip install -r requirements.txt
```

3. Starting the tool

1. Double-click **run.bat**. A window opens and stays open - that window is the program, so leave it running.
2. Your browser opens at **http://127.0.0.1:5001**.

4. File Analysis page

Upload any two, or all three, of Baseline / Before / After. Accepted formats: **.edf** files that contain real EEG channels (EMOTIV also exports a motion-only .edf with no brain signal at all - the tool rejects that one with an explanation), and **EMOTIV Pro CSV exports** named *_Original.csv or *_Cleansed.csv. Epoched workbooks (.xlsx) are not accepted here, since band power needs continuous signal.

Channels default to the occipital/parietal set (O1, O2, Oz, Pz, P3, P4), which gives the steadiest alpha reading. Tick or untick to change the selection.

5. Reading the results

5.1 The three comparisons

Section	Question answered
A. Before vs after	The alpha/beta ratio alone, before the stressor against after it.
B. Direct comparisons	Every band and both ratios, tested for each pair of recordings independently (Baseline-Before, Baseline-After, Before-After).
C. Change from baseline	Baseline treated as normal; Before and After expressed as percentage change from it. 'Stressor effect' is the After change minus the Before change - the part attributable to the task itself.

> **Why relative power, not absolute:** every cross-recording comparison uses each band's share of that epoch's total power, not its absolute microvolt-squared value. Absolute power depends on electrode contact and amplifier gain, which differ every time the cap goes on - comparing it between two separate recordings would measure how well the cap was fitted, not the participant's brain state. Shares cancel that out.

5.2 Effect size and significance

Cohen's d measures how far apart two conditions sit, in units of their own spread: 0.2 small, 0.5 medium, 0.8 or above large. The p-value is from a Mann-Whitney U test on the per-epoch values (not a paired test, since the two recordings usually run different lengths), then corrected across all six bands and two ratios tested together using the Benjamini-Hochberg procedure. Without that correction, testing eight measures at once would call about one in three 'significant' purely by chance at the $p<0.05$ threshold.

5.3 Frontal alpha asymmetry

Shown when the recording has both a left frontal site (F3, F7, AF3 or Fp1) and a right one (F4, F8, AF4 or Fp2). Computed as ln(right alpha) minus ln(left alpha): a negative value means relatively more left frontal alpha, conventionally associated with approach-oriented affect; positive means more right frontal alpha, associated with withdrawal-oriented affect. This is a descriptive index from the affective neuroscience literature, not a diagnosis.

6. Live page (Cortex API)

Requires the **EMOTIV Launcher** running and logged in, and a client ID and secret from an app registered at **emotiv.com/developer**. Enter them and press Connect. The connection follows EMOTIV's official protocol: requestAccess -> authorize -> queryHeadsets -> controlDevice -> createSession -> subscribe.

Two band sources exist, and the tool picks automatically. If your licence grants raw EEG access, bands are computed with the identical method used on files, including delta - and the live numbers are then directly comparable to a file analysis. If only EMOTIV's own band-power stream is granted, that is used instead, clearly labelled: it has no delta, and its numbers are not on the same scale as the file analysis, so do not put them in the same comparison.

> **No credential is stored by this program.** The client ID and secret are kept only in the browser tab for that session and sent straight to your local Cortex service. If Cortex refuses the connection, the page shows the exact reason (wrong credentials, app not yet approved in the Launcher, no headset paired, or a licence that does not include the 'eeg' scope) rather than a raw error.

EMOTIV's own performance metrics (stress, relaxation, engagement, excitement, interest, focus) are shown alongside when granted, clearly labelled as EMOTIV's algorithm rather than this tool's band analysis. They are a useful second opinion, not a replacement for it.

7. If something goes wrong

Message	What to do
contains no EEG / motion export	You picked EMOTIV's motion-only .edf. Use the FLEX2-named .edf or the matching _Original.csv instead.
epoched data, not continuous	.xlsx workbooks are not accepted here; use the .edf or _Original.csv for the same session.
Invalid Client Credentials (code -32021)	The client ID or secret is wrong. Copy them again from emotiv.com/developer - they are case-sensitive.
Your Cortex licence does not allow raw EEG access	Band power needs the same 'eeg' scope as raw EEG, so this affects both. Use the File Analysis page with an EmotivPRO export instead.
Could not reach Cortex	Start the EMOTIV Launcher and make sure it is running and logged in.

8. Technical notes

Setting	Value
Band edges	delta 1-4, theta 4-8, alpha 8-12, beta_low 12-16, beta_high 16-25, gamma 25-45 Hz
Why these edges	Match EMOTIV Cortex's own convention, with delta added since Cortex does not report it.
Filter	1-45 Hz, 4th order Butterworth, zero phase
Spectral estimate	Welch, matched to the epoch length, 50% overlap
Epoch length	4 s by default, adjustable on the File Analysis page
Artifact rejection	Worst-channel peak-to-peak above median + 4 x MAD
Statistics	Mann-Whitney U per measure, Benjamini-Hochberg FDR correction across 8 measures, Cohen's d on relative power
Cortex protocol	requestAccess -> authorize -> queryHeadsets -> controlDevice -> createSession -> subscribe
Compute device	CPU only

> This tool describes measured band power and its change across recordings. It is a research and self-monitoring aid, not a medical or diagnostic device.

脑电压力分析器

使用说明书 - 频段功率压力对比，支持实时与离线分析

1. 本程序测量的内容

本工具比较同一被试最多三次脑电记录之间的频段功率：静息 基线 (Baseline)、压力任务 前 (Before) 的记录，以及任务 后 (After) 的记录。它回答三个问题：三次记录中任意两次之间，所有频段有何差异；α / β 比值从任务前到任务后如何变化；以基线作为该被试的正常水平，压力任务在此基础上使每项指标额外移动了多少。

本工具报告六个频段：δ、θ、α、低 β、高 β 与 γ，并附带 α / β 与 θ / β 比值。本工具的核心指标是 α / β 比值，这是一项常用指标：数值越低，表示相对于 α 波，β 波的比重越高，通常解读为更警觉或压力更大，而信噪比一类的记录质量检查不在本工具范围内，如需该功能请使用 EEG Optimizer。

> 所有计算均在 CPU 上完成，无需显卡；文件不会离开本机。实时页面仅与本机的 EMOTIV Cortex 服务通信 (wss://localhost:6868)。

2. 安装

需要 Python 3.10 或更新版本，安装时请勾选 Add python.exe to PATH。

1. 双击本文件夹中的 install.bat。
2. 等待安装完成后关闭窗口。

```
pip install -r requirements.txt
```

3. 启动工具

1. 双击 run.bat。会打开一个窗口并保持运行 —— 该窗口即为程序本体，请勿关闭。
2. 浏览器会自动打开 http://127.0.0.1:5001。

4. 文件分析页面

上传 基线/前/后 中的任意两项或全部三项。支持格式：包含真实脑电通道的 .edf 文件（EMOTIV 也会导出仅含运动数据、没有脑信号的 .edf，本工具会拒绝该文件并说明原因）；以及名为 *_Original.csv 或 *_Cleansed.csv 的 EMOTIV Pro CSV 导出文件。此处不支持分段工作簿 (.xlsx)，因为频段功率分析需要连续信号。

通道默认选择枕叶/顶叶组合 (O1, O2, Oz, Pz, P3, P4)，该组合的 α 波读数最稳定。可勾选或取消勾选以调整选择。

5. 解读结果

5.1 三项比较

分区	回答的问题
A. 前后对比	单独展示 α / β 比值，比较压力任务前后。
B. 直接比较	所有频段及两个比值，分别检验每一对记录（基线-前、基线-后、前-后）。
C. 相对基线的变化	以基线为正常水平；前与后以相对基线的百分比变化表示。「压力效应」为「后」变化减去「前」变化 —— 即可归因于任务本身的部分。

> 为何使用相对功率而非绝对功率：所有跨记录比较均使用该频段占该时段总功率的份额，而非其绝对的微伏平方值。绝对功率取决于电极接触与放大器增益，而这些每次戴帽都会不同 —— 若在两次独立记录之间比较绝对功率，测得的将是电极帽佩戴的松紧程度，而非被试的脑部状态。使用份额可以消除这一影响。

5.2 效应量与显著性

Cohen's d 以数据自身的离散程度为单位，衡量两种状态之间的分离程度：0.2 为小，0.5 为中，0.8 及以上为大。p 值来自对逐时段数值进行的 Mann-Whitney U 检验（而非配对检验，因为两次记录的时长通常不同），随后对同时检验的六个频段及两个比值使用 Benjamini-Hochberg 方法进行校正。若不作此校正，同时检验八项指标在 $p<0.05$ 阈值下约有三分之一会仅因偶然而被判定为「显著」。

5.3 额叶 α 波不对称性

当记录同时包含左额位点 (F3、F7、AF3 或 Fp1) 与右额位点 (F4、F8、AF4 或 Fp2) 时显示该指标。计算方式为 ln(右侧 α) 减 ln(左侧 α)：负值表示左额 α 相对更强，传统上与趋近型情绪相关；正值表示右额 α 相对更强，与退避型情绪相关。这是情感神经科学文献中的描述性指标，并非诊断结论。

6. 实时页面 (Cortex API)

需要 EMOTIV Launcher 处于运行并已登录状态，以及在 emotiv.com/developer 注册应用获得的客户端 ID 与密钥。输入后点击 Connect。连接过程遵循 EMOTIV 官方协议：requestAccess -> authorize -> queryHeadsets -> controlDevice -> createSession -> subscribe。

存在两种频段数据来源，工具会自动选择。若你的授权允许访问原始脑电，频段将采用与文件分析完全相同的方法计算，包括 δ 波 —— 此时实时数值可与文件分析直接比较。若仅获得 EMOTIV 自身的频段功率数据流，则改用该数据流，并会明确标注：该数据流没有 δ 波，其数值与文件分析不在同一尺度上，请勿将两者放入同一比较。

> 本程序不会存储任何凭据。客户端 ID 与密钥仅保留在该次会话的浏览器标签页中，并直接发送至本机的 Cortex 服务。若 Cortex 拒绝连接，页面会显示具体原因（凭据错误、应用尚未在 Launcher 中获批、未配对头戴设备，或授权不包含 'eeg' 权限范围），而非原始错误信息。

若获得授权，还会一并显示 EMOTIV
自身的性能指标（压力、放松度、投入度、兴奋度、兴趣、专注度），并明确标注为 EMOTIV
算法的结果，而非本工具的频段分析。它们可作为有用的参考，但不能替代频段分析。

7. 常见问题处理

提示信息	处理方法
不含脑电数据 / 运动导出文件	选择了 EMOTIV 的运动数据 .edf 文件，请改用带 FLEX2 名称的 .edf 或对应的 _Original.csv。
分段数据，非连续记录	此处不支持 .xlsx 工作簿，请使用该场次对应的 .edf 或 _Original.csv。
凭据无效 (代码 -32021)	客户端 ID 或密钥有误。请重新从 emotiv.com/developer 复制，注意区分大小写。
当前 Cortex 授权不允许访问原始脑电	频段功率与原始脑电需要相同的 'eeg' 权限范围，因此两者都会受影响。请改用文件分析页面配合 EmotivPRO 导出文件。
无法连接 Cortex	请启动 EMOTIV Launcher，并确认其处于运行且已登录状态。

8. 技术说明

参数	取值
频段划分	delta 1-4, theta 4-8, alpha 8-12, beta_low 12-16, beta_high 16-25, gamma 25-45 Hz
选择依据	与 EMOTIV Cortex 自身的划分一致，并加入 δ 波，因为 Cortex 本身不提供该频段。
滤波	1-45 Hz，4 阶巴特沃斯，零相位
谱估计	Welch 法，窗口与分析时段等长，50% 重叠
分析时段长度	默认 4 秒，可在文件分析页面调整
伪迹剔除	最差通道峰峰值超过 中位数 + 4 x MAD
统计方法	各指标使用 Mann-Whitney U 检验，8 项指标间使用 Benjamini-Hochberg 方法校正 FDR，并对相对功率计算 Cohen's d
Cortex 协议	requestAccess -> authorize -> queryHeadsets -> controlDevice -> createSession -> subscribe
计算设备	仅使用 CPU

本工具描述所测得的频段功率及其在不同记录间的变化，属于研究与自我监测辅助工具，并非医疗或诊断设备。